

2 – Investigation

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2.1 Investigation into Current System

2.1.1 Investigation

Methods of Investigation	Justification
Observation	This allows me to observe how the system is actually operating in practice and I will be able to identify problems as they arise. It also means that I can verify and ensure that the findings from the other methods of investigation are free from bias and are accurate.
Questionnaire	Using a questionnaire, I can gather a larger number of responses than what is possible with the other methods of investigation. This allows me to see a broader picture of what all the users want in a new system and what they dislike about the current system. Because the questionnaire is anonymous, it's more likely that I will get honest responses from it compared to an interview, for example.
Interview	During an Interview, I can ask precise questions and gather a large amount of detailed information from a single interviewee; this would be Steph. I can also ask follow up questions to gather more detail if needed. This allows me to more thoroughly investigate certain areas of the current system and find out exactly what Steph is looking for in a new system.
Document Analysis	This allows me to see how Steph currently manages the data that she stores and what exactly is stored. It would be more time consuming to gather this information by using any of the other methods of investigation and there is the potential that Steph may forget some data that she stores if she told me verbally or made a list of it.

2.12 Discussion and Key Findings

Observation

I attended a computer club session and observed for the entire session. I turned up before the session started and left after it ended to ensure that I would have a complete picture of how the sessions ran and the problems encountered, as summarised below:

- The learnmyway.com system wasn't able to be used because there was no internet access. A mock-up form that Steph had made using Word with some interactive features had to be used.
- The users weren't able to have the experience of logging into a website and locating a form as the form was just a Word document on the desktop so Steph explained that in the real world they would have to log into the website.
- Many users struggled to find the start of the form as they had to slowly scroll down the document to find it and many users accidentally scrolled past it.
- It would be useful for the system to be able to fill the address automatically when the user enters their postcode as Steph had to explain to them that the address would automatically fill when they're filling out a real form. One of the users was confused before this explanation and kept repeatedly entering their postcode, expecting it to automatically fill the information.
- Due to the limitations of word documents, the forms weren't very accurate. Some of the form elements that should have been radio buttons were checkboxes and this confused most of the users.
- There are accessibility issues with the current system. One of the users was unable to read the text on the screen as they had very poor eyesight and so a volunteer had to sit next to them and read everything out for them.
- Some of the users were more proficient than others at filling in the form and completed it much earlier. This meant that they had to wait with nothing else to do as they had finished their form and no further forms were available.
- There was no validation or double entry verification on the form therefore some users had incorrectly filled the forms in and were unaware that they had filled them in wrong.
- Some elements of the forms weren't interactive. Some of the checkboxes on the forms weren't actually checkboxes and were just squares and this was confusing for the users.
- Because Word wasn't designed for mocking up forms, the document was very complex and Word struggled to display it. A user's instance of Word crashed and they lost all their progress on the form.

Questionnaire

I created a questionnaire using Google Forms with 8 questions and distributed it to the users/members of the group. I tried to make the questionnaire as simple to answer and understand as possible by making all of the questions agree or disagree questions. This is because the members of the group are currently learning how to fill in online forms and aren't very proficient in filling them in. This means that it's most likely they would struggle to fill in a complicated form.

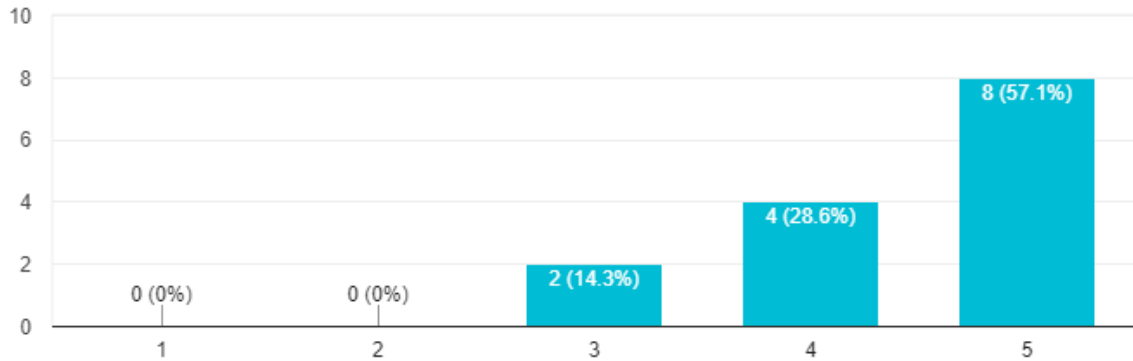
Although I created it as an online form, as the group currently has no internet access, I provided Steph with a pdf of the questions. She printed off the forms and the users filled them out on paper.

There were 14 responses in total.

For each question, the user could respond on a scale from 1 to 5, where 1 is strongly disagree and 5 is strongly agree.

It would be useful to be able to create forms myself by selecting questions from a list

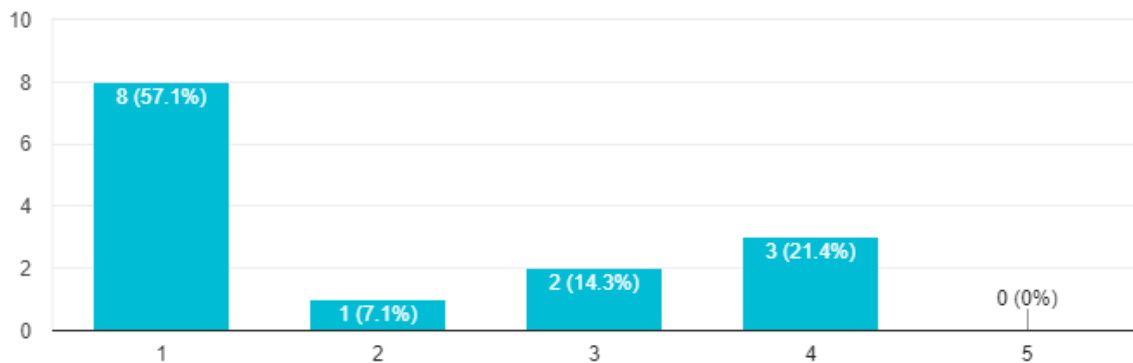
14 responses



As you can see from this, the vast majority of the users agreed to “It would be useful to be able to create forms myself by selecting questions from a list” This means that this feature should be priority for the new system and should be able to be easily used by the users as it’s a feature that almost all of them want.

There is a wide number and range of forms and questions available for me to practice

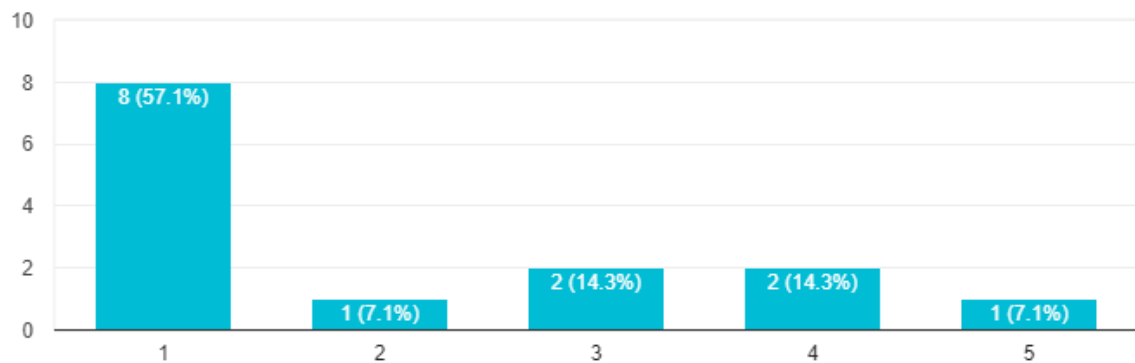
14 responses



There was a greater range of responses to this question. This is probably because the wording “there is a wide range” is ambiguous because “a wide range” isn’t quantitative. The majority of the users strongly disagreed to this question. This has highlighted that the number and range of forms in the system needs to be improved. Based on the responses to the previous question it’s likely that the users would like to be able to do this by creating the forms themselves.

I can easily search for and find a form to practice in the system

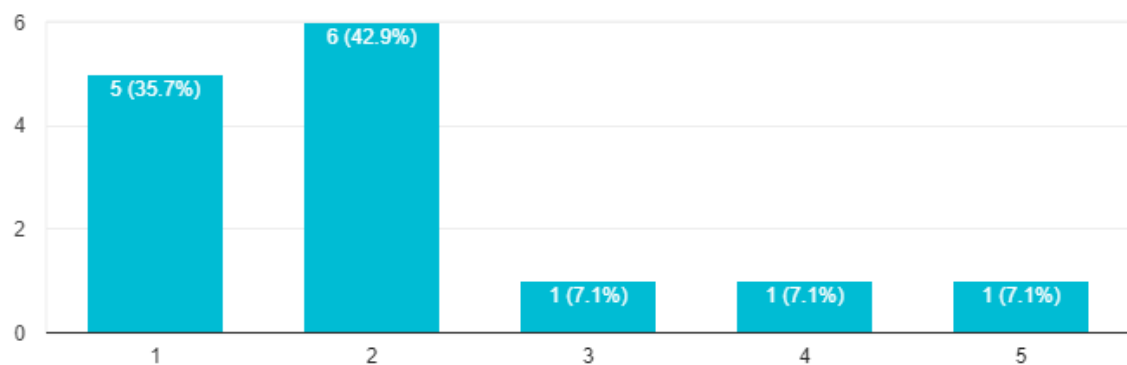
14 responses



This is another question with a wide range of answers. It's most likely that the range is due to the differences in each user's skill, as the more advanced users will find this task easier than the less advanced users. The majority of the users strongly disagreed to this question. It's important for the new system to improve on the current methods of finding form in the system (looking through the file explorer) to make it more accessible to less advanced users, who struggle with the current system.

I can easily see where I've made an error filling in a form because the system tells me and highlights the error

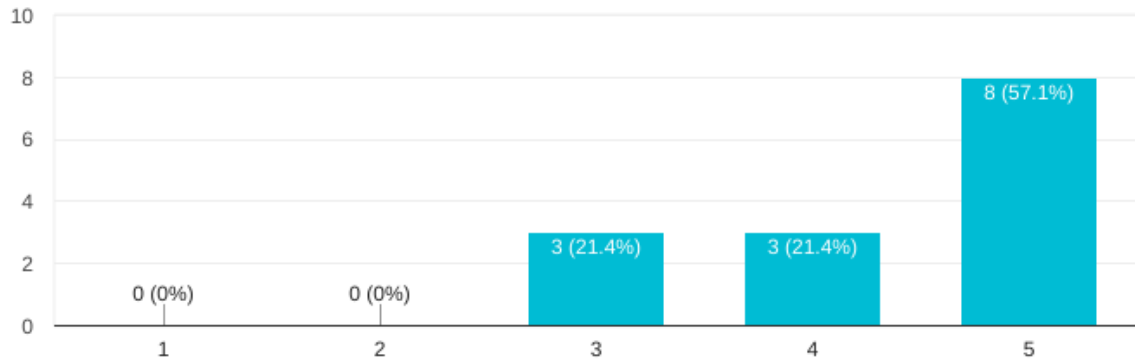
14 responses



The vast majority of users disagreed to this question. This shows that the current system has a large problem with highlighting errors. This is also reinforced by what Steph has said in the interview.

It would be useful to see what questions I commonly struggle with and which ones take me the longest to complete

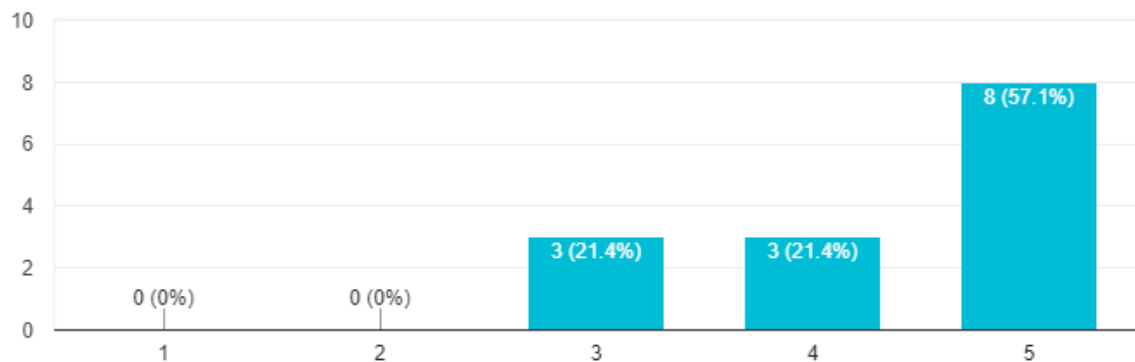
14 responses



No users disagreed to this question and the majority strongly agreed. This shows that the users would find it useful for the system to tell them which questions they need to improve on. In the interview, Steph said that she would like to be able to track progress like this but currently doesn't have the capability. This is a feature that both Steph and the users want therefore it's important for it to be included in the system.

The current system is confusing to use

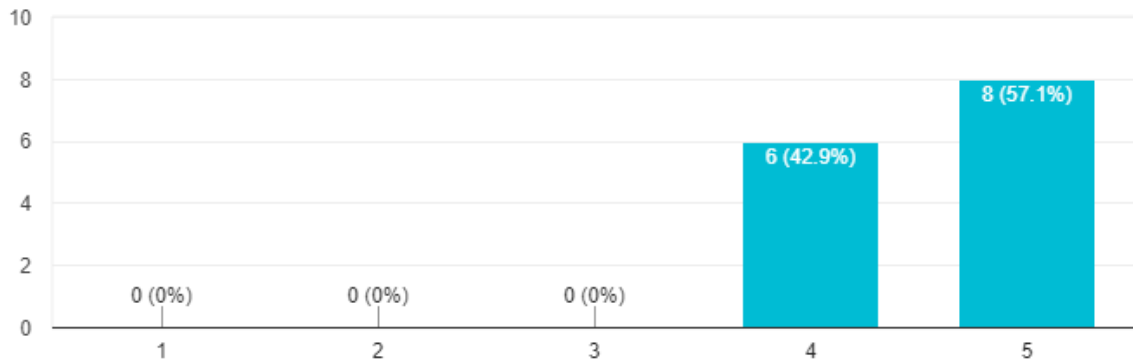
14 responses



Almost all of the users agreed that the current system is confusing to use. This is a key area that should be improved in the new system as it's very important for the system to be easy to use because the users have limited IT skills.

A help menu which explains how to use each function of the system would be useful

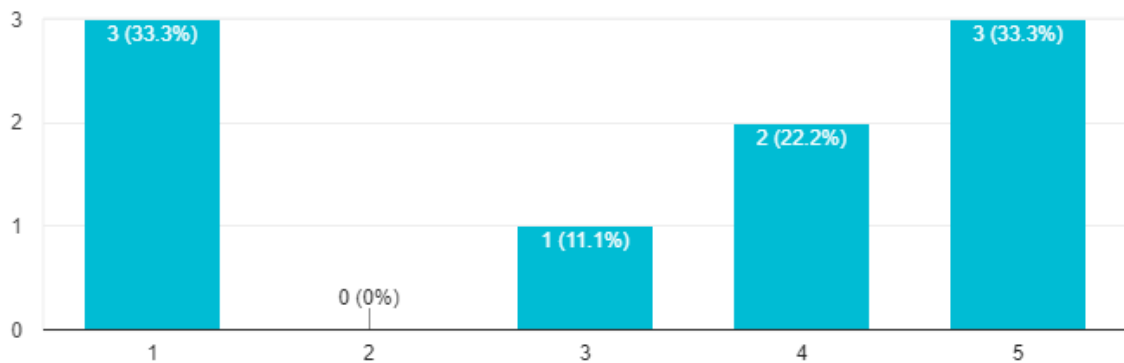
14 responses



All users agreed to this question with the majority strongly agreeing. This feature of the new system would help the new system be less confusing for the users which is important because the majority of them found the old system confusing. Therefore, this is an important feature that should be included in the new system.

The forms I've practiced are realistic and are similar to the forms that I've encountered in the real world. (Only an... if you've completed a real world form)

9 responses



This was the final question of the form. This question has fewer responses as it wasn't a required question.

The users were very split on this question. The users who agreed to the question will have most likely filled in a blue badge form in the real world and this is a form that they have extensively practised at the computer club. Other users may have filled in a completely different form in the real world that isn't familiar to them and so will have disagreed to this question. This makes this data difficult to draw conclusions from because the question didn't ask the users for reasons why the forms weren't/were like those that they've encountered in the real world.

Because of this split, I will be mostly ignoring the responses to this question.

Interview

I interviewed Steph twice - once over the phone and once through text messages.

Phone interview summary:

- A realistic (true to the real world) way of finding and searching for forms isn't important.
- When using the word documents, there is no way of knowing if the form has been completed correctly. Steph, or another volunteer, has to manually look through the form to see if the user has filled in something wrong.
- Neither Learnmyway.com nor the word documents mimic the real world very well and this is a key area for improvement.
- Steph likes that the current system is very plain speaking and easy to understand, that there is an audio or text option to read/listen to the instructions. She also likes that it doesn't assume any prior knowledge - it's for complete beginners.
- The main problems with learn my way are:
 - Users aren't able to start the forms at the very beginning as it starts them part way through the form with most of it filled in.
 - They don't have to create an account.
 - There is no submitting the form as you only have to answer one or two questions.
- The main things that Steph is looking for in a new system are:
 - The ability to start a form from the very beginning and fill it in up to completion and submit it.
 - A range of different types of questions e.g. drop downs, date entry via a calendar etc.
 - The ability to upload photographs.
 - The form won't let you submit it if there are any errors. It will also highlight what's wrong like what happens in the real world.
- Steph keeps a register and she stores information such as the users' phone numbers.
- Nothing about the users' progress is stored because currently they don't have the capability for it. But Steph would really like to be able to.
- The ability for the system to give a small tutorial on each type of question isn't of much interest. Steph would want the tutorial to be as small as possible because some of the users have early onset dementia and learn better by doing rather than reading the tutorial.
- A simple way of creating new forms by selecting questions is a feature that Steph would like. She also likes the idea of the system being able to automatically create forms based on user weaknesses.
- The progress metrics that Steph would like tracked are:
 - Time taken to complete a question
 - Number of times that a user has failed a validation check
 - How many times it's taken a user to correct an error.

Text Interview

- Attendance, phone numbers, date of birth, email addresses and passwords (for Learnmyway.com) are stored.
- Steph stores the data confidentially on her laptop in an access database.
- Steph stores the attendance in case they don't see a user for a while so she can look back through the records and see if they've been attending for the past few weeks.

- Steph stores the email addresses and passwords of the users for the learnmyway.com system because the users often forget them. She only stores the email addresses and passwords of the users that she thinks are likely to forget them.
- Steph stores paper attendance logs and the names and numbers of new users in a ring binder and then copies them over to the access database.
 - Steph takes the names and numbers verbally from the users
- When a user wants to know their username and password Steph searches through the database by name in order to find it.
- Steph sometimes sorts the database alphabetically by name and filters by who attended the most recent session.

Document Analysis

Access Database

Steph has provided me access to the Microsoft Access database that she uses to store all of the information about the users and the group.

There are 9 tables:

- **Achievements**

#	Field	Field Description
1	Id	Used to uniquely identify each achievement and so allows each volunteer to easily have achievements assigned to them and avoids the redundancy of writing out the full achievement names for each volunteer.
2	Achievement name	The name of the achievement e.g. 5-9 years of service

This table contains a list of the achievements a volunteer can be awarded with. It has no other data in it and is linked to by other tables.

- **Attendees**

#	Field	Field Description
1	Id	Used to uniquely identify each attendee to form links between the various tables.
2	Attendee	Contains both the first and last name of the attendee
3	Address	Where the attendee lives.
4	Phone Number	The attendee's home/mobile number
5	Email Address	The attendee's email address that they use to login to learnmyway.com in case they forget it
6	Password	The attendee's password that they use to login to learnmyway.com in case they forget it

A list of all the users that attend the sessions. Every user that has ever attended is in this table even if they don't regularly attend. Not every field is filled out for every user. For example, the password and email address fields. This is because, as Steph said during the interview, she

only stores them for the users that she thinks are likely to forget them. Id and attendee are filled out for every user.

- **Course names**

#	Field	Field Description
1	Id	Used to uniquely identify each course and so avoids the redundancy of writing out the full course name multiple times.
2	Course name	The name of the course e.g. Starting Online

A list of the different courses that are run at the computer group e.g. Online basics. This is similar to the achievements table as it seems to just exist to be linked to by other tables to avoid redundancy.

- **Meeting Attendees**

#	Field	Field Description
1	Id	Used to uniquely identify each meeting each attendee has attended.
2	meetingid	This is linked to the meetings table so rather than the actual id of the meeting being displayed the date of the meeting is displayed.
3	attendeeid	The attendee. This is linked to the attendees table and so, rather than the id of the attendee being displayed, the name of the attendee is displayed.

Stores which volunteer/user went to which meeting. This table is poorly laid out. There is a row for each meeting each user attended. For example, there are around 30 rows for just the meetings that Steph attended. However, there is an advantage to this layout as it makes searching by user and by meeting easier as the data isn't grouped - the raw data is available.

- **Meetings**

#	Field	Field Description
1	Id	Used to uniquely identify each meeting and is used to form links between the tables. E.g. this table is linked to the meeting attendees table
2	meetingdate	The date of the meeting. Written as dd/mm/yyyy
3	status	Whether the meeting was Active, Drop-in, or Cancelled
4	Week number	The number of the week that the meeting was held in. It's unclear where the counting started and all the week numbers are between 0 and 10
5	Volunteer Att	The number of volunteers that attended the meeting
6	Friends	The number of users/non-volunteers that attended the meeting

7	Subs raised	The amount of money raised through subs
8	Tea/coffee	The amount of money raised from selling tea and coffee
9	Costs	The amount of money spent
10	Sales	Money raised from things other than subs and tea/coffee. E.g. selling cards that they've made
11	Amount raised	The total amount raised / lost in the meeting
12	Notes	Some notes on where the money has come from / been spent

This stores every meeting that has been held. The fields other than id, meetingdate, and status are only filled in for the meetings after July in 2016 so it seems like these extra fields were only added at that time. The week number field is sparsely filled in. The 5 fields that deal with money appear to be used as a basic method of bookkeeping. There is also a plus button on the far left of each row that, when pressed, shows a small extract from the meeting attendees table showing who was present at that meeting.

- **Training**

#	Field	Field Description
1	Id	-
2	Attendee	-
3	Volunteer	-
4	Course	-
5	Date Started	-
6	Date finished	-
7	LMW Username	-
8	LMW Password	-
9	Email	-
10	Email Password	-

This table appears to be for tracking the progress of volunteers/users however there is no data in it at all therefore it isn't currently used.

- **Volunteer Achievements**

#	Field	Field Description
1	id	Used to uniquely identify each row in the table
2	volunteer	The volunteer that the achievement/row corresponds to
3	achievement	The achievement that was achieved. This is an achievement from the achievements table and is linked here .
4	Date Achieved	The date that the achievement was achieved stored as dd/mm/yyyy

5	Management	A checkbox that is ticked if the volunteer that the row corresponds to is in management.
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This stores the achievements that each volunteer has achieved and the date that they achieved them. Each volunteer has a row for each achievement that they have achieved, therefore each volunteer may be in the table multiple times.

● Volunteers

#	Field	Field Description
1	Id	Used to uniquely identify each volunteer
2	FullName	The full name of the volunteer is stored in this field so that each volunteer can be easily identified by name
3	Address	The home address of the volunteer
4	Phone number	The home phone number of the volunteer
5	Mobile number	The mobile number of the volunteer
6	Email Address	The email address of the volunteer
7	Date started	The date that the volunteer started volunteering. Stored so that the years service achievements can be easily awarded.
8	Active	Ticked if the volunteer is active / regularly attends meetings.
9	Up to 4 years	Ticked if the volunteer has achieved up to 4 years of service.
10	5 - 9 years	Ticked if the volunteer has achieved between 5 and 9 years of service
11	10-19 years	Ticked if the volunteer has achieved between 10 and 19 years of service
12	20-24 years	Ticked if the volunteer has achieved between 20 and 24 years of service
13	Management	Ticked if the volunteer is in a management position
14	Occasional	Ticked if the volunteer is only an occasional helper

This table contains a list of all the volunteers. The full name and date started fields are filled in for every volunteer. The address, phone number, mobile number, and email address fields are only filled in for 2 volunteers out of the 64 in the table. There are links between the volunteers table and the volunteer achievements table. If you click a plus button on the far left of each row, it'll show an extract of the achievements table that corresponds with the volunteer. This shows the same years of service achievements that are present as check boxes in the volunteers table and also shows what other achievements from the achievements table they have achieved e.g. "Computer Group Certificate"

● Volunteer Training

#	Field	Field Description
1	Id	-
2	Volunteer	-
3	Training Course	-

4	Date completed	-
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This table is another table that isn't filled in at all, and so presumably it's not used either.

Paper Register

#	Field	Field Description
1	Date	The date of the meeting
2	Name	The name of the attendee - usually just the first name
3	Time in	The time that the attendee arrived
4	Time out	The time that the attendee left

Steph has provided me with some photographs of the paper register that she uses. The register is laid out in a table. The date and name fields are filled out for every attendee. There is a new line of the table for each attendee of each meeting. The time in and out columns are only filled in for a few attendees. This is possibly because those attendees are in management positions or are the few members of Halliwell Befriending Service that are paid so it's important to keep track of what time they arrived and left. I will not be storing the time in and out in my system as it's unnecessary for the regular attendees of the group.

Steph distinguishes between different meetings in the paper register by putting a line between each meeting. It appears that she reuses this register each week.

2.14 Inputs, Processes & Outputs of Current System

Forms

Inputs	Description	Example
Title	The title of the person filling the form in.	Mr.
First Name	The first name of the person filling the form in	Bob
Surname	The surname of the person filling the form in	Smith
Gender	The gender of the person filling the form in	Male
Date of birth	The date of birth of the person filling the form in	06/07/1996
Town of birth	Which town the person filling the form in was born in	Bolton
Country of birth	Which country the person filling the form in was born in	England
National Insurance Number	The national insurance number of the person filling the form in so that their identity can be verified	QQ 12 34 56 C
Address details	The address of the person filling the form in, needed for payments or deliveries	10 A Street A Town YU2 9XD
Telephone number	The telephone number of the person filling the form in so that they can be contacted	01204 123456
Email address	The email address of the person filling in the form so that they can be contacted via email	example@example.com
Identification photo	A passport style photo of the person filling the form in to be uploaded	

Processes	Description
Search forms to distribute to the users	Steph has to manually search through her hard drive to find the particular form that she wants the users to practice.
Validation / searching for errors	Because the form isn't automatically validated the volunteers have to manually look through the form that the user has filled out to look for errors.
Copying of forms	Steph has to copy the form that she wants the users to practice to a USB stick and then copy it to all the laptops in the club.
User searching for form to completed	The users have to search the desktops on their laptops for the form that Steph has previously copied there and must open it in order for them to begin completing it.
Search for start point of the form	The user has to manually search through the word document that contains the form in order to find the start point as the form doesn't automatically open at the correct start point.
Saving progress	When the session comes to an end the users need to save the form that they're working on. To do this, they open the save file menu of word, search for their USB stick in file explorer, give the form a name, and save it.
Opening previously/partially completed forms	If the user wants to open a form that they've previously/partially completed they need to manually search their desktop/USB stick until they find the form.

Outputs	Description
Alert user that they've made an error.	The volunteer verbally tells the user if they've made an error after they've looked through the form that the user has filled in.
(Partially/Fully)Completed form	The user can save the form and open it at a later date.

Progress report	The volunteers can verbally tell each user how they're progressing and where they need to improve. This is to aid their progress and improvement.
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Digital Register / User Database

Inputs	Description	Example
Forename	Allows users to be identified by first name in the database	Janet
Surname	Allows users to be identified by last name in the database	Fletcher
Mobile Number	Allows users to be contacted	0771 2345678
Phone Number	Allows users to be contacted if their mobile isn't available / can't be reached	01204 123456
Email Address	Stored in case the user forgets their email address	bob@outlook.com
Password	Stored in case the user forgets their password	password
Address	Enables Steph to see where each user lives	1 A street BL17JH

Processes	Description
Add User	Steph has to manually fill out a row in the database with the new user's details. The new user's details are taken from the user verbally.
Remove User	If a user leaves the club, Steph can delete their corresponding row in the database
Search for user by ID	The database can be searched for a particular user by ID. This can be done to find which sessions a user was present at.
Filter by session	The meetings table can be filtered by individual meeting/session to show who was present.
Search for session by date	The meetings table can be searched to find the meeting/session for a specific date to show who was present.
Search for session by year	The meetings table can be searched to find all of the meetings/sessions that ran during a particular year.
Add volunteer	Steph can add a volunteer by manually filling out a row in the database.
Remove volunteer	If a volunteer leaves the club, Steph can delete their corresponding row in the database
Filter by active volunteers	The database can be queried to find all the active volunteers

Outputs	Description
Attendance report	A report can be produced for each session which shows a list of the users and volunteers that were present. This makes it clear to Steph which users were present at each session without having to look through the registers.
Attendance report for a specific meeting	A report can be produced which shows a single session and which users and volunteers were present.
Attendance report for a year	A report can be produced which shows all sessions for a particular year and which users and volunteers were

	present.
List of active volunteers	The queried list of active volunteers can be viewed and printed

Paper register

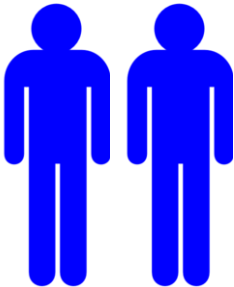
Inputs	Description	Example
Forename	Allows the user to be found in the digital database when copying up the register	Janet
Surname	Allows the user to be found in the digital database when copying up the register	Fletcher
Present	A tick or cross depending on whether the user was present or not	X
Date	The date of the session that the register was taken	01/02/18

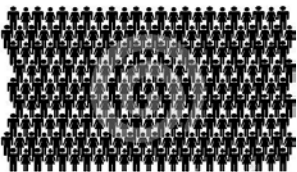
Processes	Description
Add user	To add a user to the register Steph must write their names and put a tick or cross whether they were present.
Remove User	If Steph has made a mistake or wants to remove a user from the register she must cross out what she has written about that user.
Save / store register	Steph puts the register in a binder with the other registers to store it.
Search for user	In order to find a specific user in the register, Steph must manually look down the list of users in the register in order to find the user.
Sort by date	In order to sort the registers by date Steph must take each register out of the binder and manually compare dates until they are in order. This is done periodically to reorganise the binder to make it easier to search through.
Search for register	Steph must manually look through each register in the binder until she finds the register. She will usually do this to find the registers that she has not yet added to the access database.
Produce digital register	Steph looks through each paper register and copies it up into the access database so that it can be accessed digitally and more easily.

Outputs	Description
Digital register	A digital version of the paper register stored in the access database for easy searching and report generation.

2.15 Stakeholders Involvement in Current System

Group leader (Steph)	Description
	● Steph runs and manages the group this involves: ○ Planning the lessons ○ Making the practice forms. ○ Delivering the majority of the general teaching to the users. ○ Tracking the attendance of each user and storing personal data about them. ○ Storing and managing the email addresses and passwords of each user which are used to log in to the online system.

Volunteers	Description
	● There are a few volunteers that help out at the group. They aid Steph in teaching the users how to fill in the forms and use the computers. ● Some volunteers sit with a specific user and give them 1:1 help throughout the session. ● Other volunteers sit more centrally so that they can help any of the users if they require help. ● The volunteers usually don't have to directly use the system however they often have to verbally walk the users through completing the form. ● They also look over the forms that each user has filled in to determine if the user has made any errors. ● There are a few regular volunteers that help out every week. However, there are some others that help out whenever they can and not regularly.

Users	Description
	● The users of the system are the elderly people that attend the club. ● They use the system to improve their skills of filling in forms by practicing. ● These are the main stakeholders in the system as the entire system exists to help them learn these skills.

2.16 Limitations of Current System

Limitation	Consideration
Some of the forms lack validation.	This is an essential feature of the system because, without it, the users wouldn't know if they're filling the forms in wrong. As this feature is lacking in the current system, the volunteers have to constantly check how each user is doing and have to check their answers to the form manually to ensure that they're correct. This is time consuming for the volunteers. It's also unrealistic for the forms to not have automatic validation.
There is no way of tracking a user's progress and identifying their weaknesses	The current system doesn't track each user's progress. This means that it's difficult to see if a user is improving. If the system was able to identify the types of questions and forms that each user struggled on the volunteers would be able to see what each user needed help with at a glance and so would be able to give more effective help immediately. Currently, the only opportunity for a user to ask for help is when they encounter a question that they can't do. Many users struggle to explain which questions they have struggled with in the past as they aren't familiar with the names of the different types of questions. The system tracking each user's weaknesses automatically would eliminate these issues.
There is no way to quickly and easily create forms.	Currently, if Steph wants to create a form, she has to do it using Microsoft Word. This requires her to spend time manually adding each component to the form and positioning it. This is difficult because Word was not designed to create forms and, even after creating the form, there is no validation so it's not an incredibly useful or realistic form. It would save a large amount of time if she was able to just select multiple questions and the system would create the form automatically. This would also mean that she would be able to create a greater number of forms therefore the users would have a wider range and number of forms to practice.
More advanced features like automatically completing addresses and uploading/attaching documents aren't available.	Because many of the forms are created in Microsoft Word, there is no way to allow the users to upload / attach a document. Using the file explorer is a key skill that many users need to practise therefore, because they are not able to practise, they will struggle do it when they don't have the help of the volunteers. Uploading and attaching documents is a large element of many important online forms therefore it would be very useful to the users if they were able to practise it. Another feature that these forms created in Word lack is the ability to automatically complete addresses from a postcode. Many users struggle and are confused when their address isn't provided for them to select from a dropdown after they enter their postcode. The form would be more realistic if it automatically completed addresses and therefore would prepare the users better for real world forms.
Very small range and number of forms	The current system only offers forms that Steph has created herself for the users to fill in. The users can either fill these forms in or practise a specific question type using Learnmyway.com however, there are usually only 2 practice questions for each type. Because it's so time consuming for Steph to create forms, there is a limited number and range of forms that she has created. This means that the user often has to practice filling in forms that they have already filled in before. While they will still be able to improve their skills by doing this, it would be more useful for the them to attempt filling in new and different forms. This will test them more as they will have to pay full attention to the form and read it to understand what it is asking for rather than just remembering from their previous experience with the form. This better replicates what they would have to do when filling out a real word form.

2.2 Research into Existing Solutions

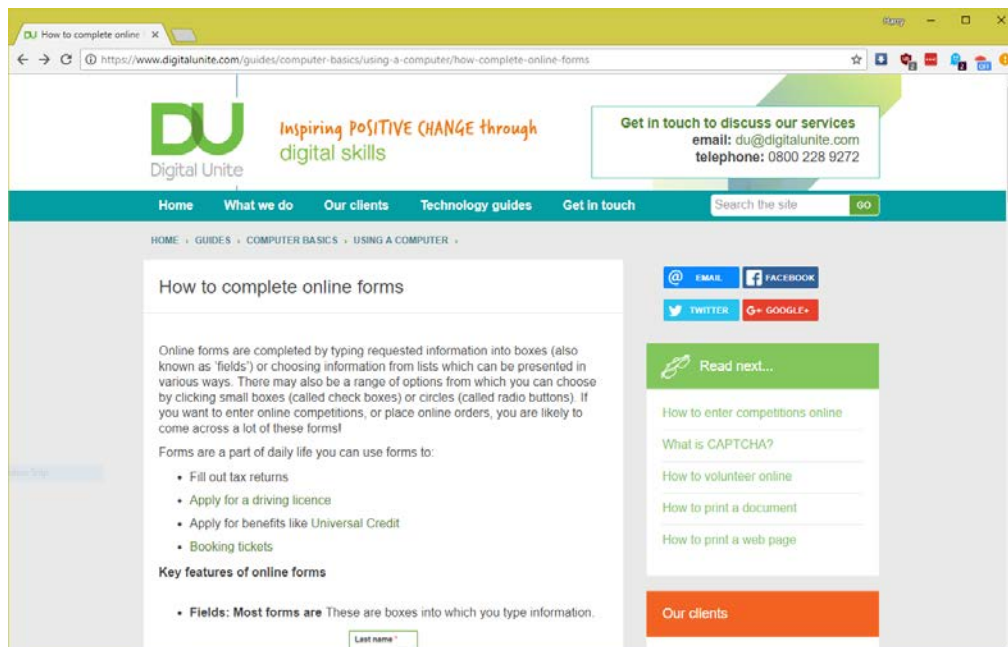
The system that I'm going to develop is very specialised and unique. Because of this, there are little/no existing solutions that fulfil the same purpose and meet the requirements for the system.

I've researched in to these existing systems:

- Digital Unite
- Official DVSA Theory Test Kit App
- Google forms

I've chosen 'Digital Unite' because it aims to help people improve their computer skills (and this is similar to the aim of the computer club), and I've chosen the 'Official DVSA Theory Test Kit App' to analyse its progress tracking features because the system that I am going to develop needs to have good progress tracking. I've chosen 'Google Forms' because although it doesn't teach users how to fill in forms it provides a facility to create forms. Creating forms is an important aspect of the system.

Digital Unite



Digital Unite is a company that provides support for many people to learn how to use computers and develop 'digital skills'. They do this by training people that are staff (in different organisations), are residents (of a care home for example) or volunteers and turning them into what they call 'Digital Champions'. They provide a service called 'The Digital Champions Network'. This is what is used to train the 'Digital Champions'.

These Digital champions then go on to train and help others within their own organization.

The website also offers many different text tutorials on how to use computers that beginners can read. I will be looking at these.

The tutorials are organised into 12 categories. There are only a few tutorials that are aimed at helping people to fill in online forms. I think that this aspect of the system is lacking because filling in forms is an essential part of using the computer and so the service may not fully prepare the users as they only have very few tutorials. My system will have many more forms that the user can practice.



The tutorial called “How to complete online forms” is easily accessible in the “Computer Basics” category under “Using a computer”. I like that the system is easy to navigate. This is especially important for the type of users that will be using the system.

The tutorial begins by telling the user that online forms are usually completed by typing requested information into boxes.

Further down the page, it shows pictures of each type of data entry with a short explanation of how to use it.

- **Radio buttons:** These are small circles which you click to select. The one you've clicked will have a dot in it.

A screenshot of a web form showing two radio buttons. The first is labeled 'Male' and is unselected. The second is labeled 'Female' and is selected, indicated by a small black dot inside the circle.

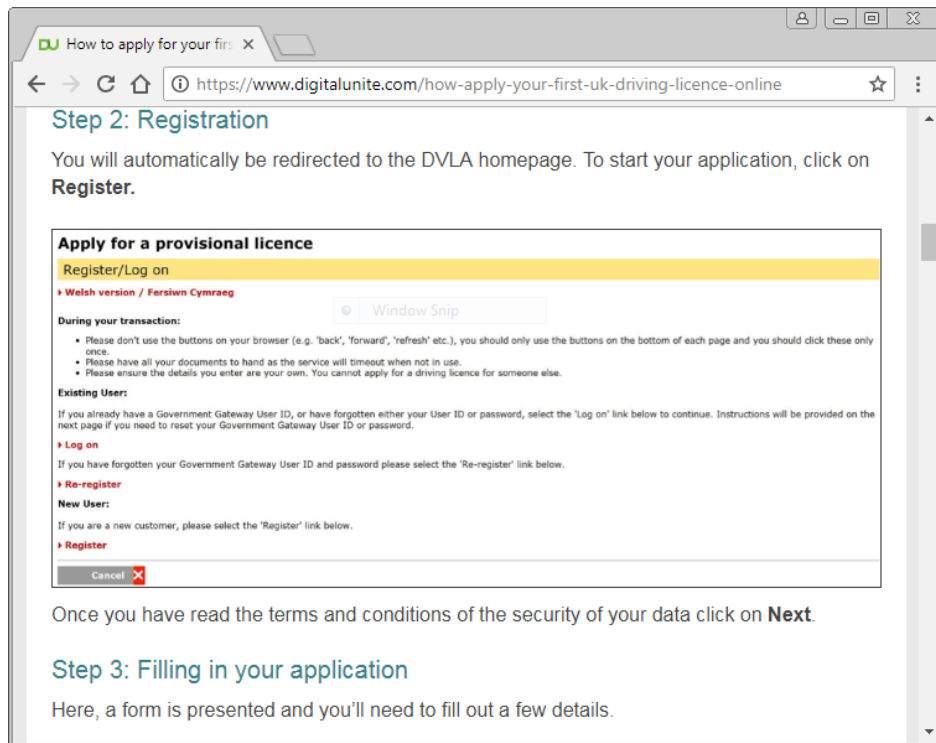
I think that the explanations for each type of entry field are very clear and easy to understand and that having the explanation with a picture of the field is very helpful to help the users learn how to fill in online forms. However, I think that this section could be improved by adding actual functioning versions of the fields for the users to interact with rather than just pictures of them.

Nearer the bottom of the page, there is a list titled “Important things to remember”. This list contains some information on things that the user is likely to encounter. For example:

- Don't press the Enter key until you've finished completing the form. If you press it earlier, it may think you've finished which means that you will probably have to start again from the beginning.

I think that these tips are very useful and mean that the users would make fewer small/simple errors.

Digital Unite also has a few full walkthroughs of common online forms. Such as this one for the DVLA form to apply for a provisional driving license.



This is really useful as it means that users know exactly how to fill in particular online forms. This is a great supplement to the walkthroughs on how to fill in each question type as it shows how to approach a full form and it's also a real form that the user may encounter rather than just a practice one that Digital Unite has created themselves. I like this feature of the system however I won't be adding it to my system as it would be too time consuming to implement.

Official DVSA Theory Test Kit App



This is an app developed by TSO (The stationery office) on behalf of the DVSA in order to help people pass their driving theory test. Although this app doesn't deal with online forms, and is teaching a different skill entirely, there are still various things that are relevant to the system that I am developing. The main feature of the app that I will be looking at is the progress tracking.

When the app is opened, there are two options, Hazard Perception Practice, and Theory Test Practice. The image on the right shows what happens when you press Theory Test Practice. I don't like how the app is split into two completely distinct and separate sections I think that it is confusing to have to pick between them when you open the app rather than the app opening into a screen like the one shown on the right. I think that these two sections should be separate apps.

On the screen shown on the right, there is a progress bar showing overall progress and, below it, there are two percentages and an average score. The first percentage tells the user how much of the study material they have read. The middle percentage tells the user what percentage of the practice questions for each topic they've completed and the average score tells them the average score that they have achieved across all of the mock tests that they've completed.



I think that this layout makes it really clear for the user to see how they're progressing at a glance but it's a little unclear what the progress bar is actually showing.

I'll be looking at the practice areas and the mock test areas. I will not be looking at the study area because my program will be focused on allowing the users to practice filling in forms and not teaching them how to fill in forms.



Shown on the left, once the user has selected practice, they are able to see a list of the all the topics. Each topic is coloured either red or green, depending on what percentage of the practice questions they have completed.

There is also more detailed information about the user's performance in each topic listed.

From this list, the user can easily select which topics they want to practice and there is a large and clearly labelled start button. Once the user presses this button, a practice test is automatically created for them to attempt. Once they have completed the test, their score is saved and used to update the information in this screen and the progress tracking screen.

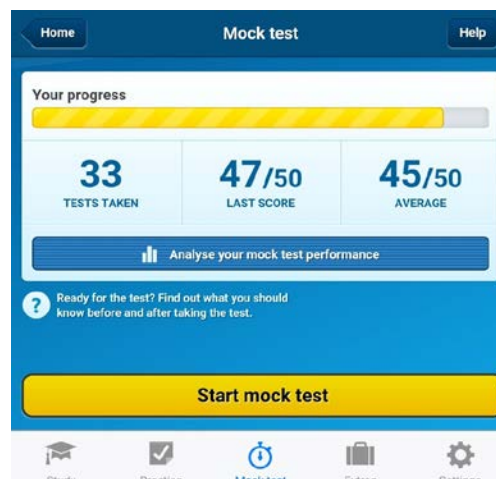
I think that this a good way to allow the user to practice their weak areas because it makes it very clear what topics they're struggling on and allows them to create a practice test from those areas very quickly and easily.

The information about progress shown on this screen is the same information that this shown on the dedicated progress tracking screens in the app so I will not be showing those screens.

I think that the information shown in this screen is a little lacking as it only shows the percentage of the practice questions that are complete. It would be more useful if it showed how often the user gets the question wrong and

how long it takes the user to complete each question. These are things that I will be improving on in my system.

If the user selects the 'Mock Test' option (by pressing the displayed average score) the screen shown below opens.



The user can clearly see their most recent score, how many tests they've taken, and their average score. Like the practice screen, there is a clearly labelled and distinct button which can be pressed to start the mock test. There is also a help button in the top right of the app. If the user presses this button, some information is displayed on how to use the mock test function such as how to pause the test and how to flag a question so that they can come back to it later. There is also information on what the actual test is like.

I think that this is good layout because it means that the user can easily see how they're progressing. The distinct start test button and the blue panel containing all of the progress analytics means that this screen of the app is well organized and not cluttered this makes this section of the app easy to use. I think that the progress bar could be improved as it's not really clear what it's for and what it's showing.

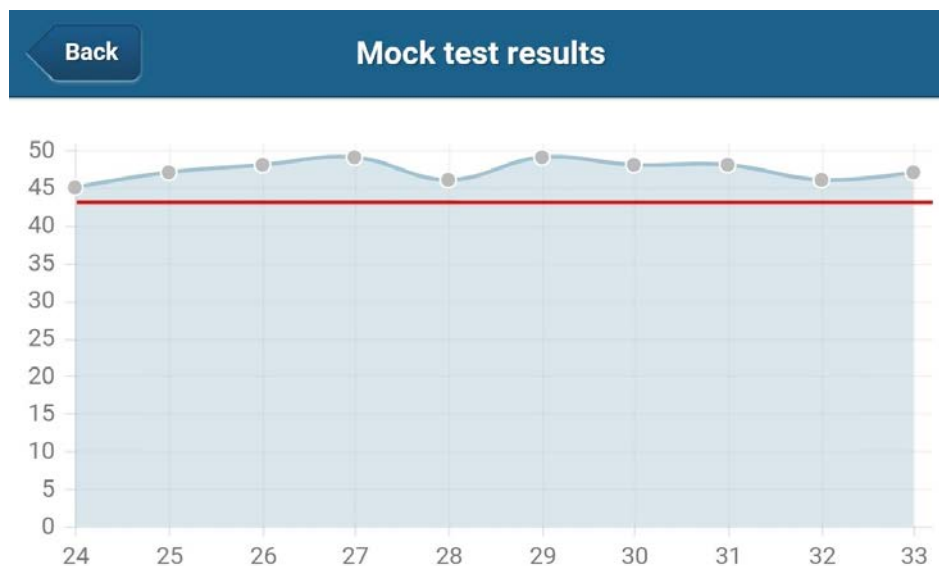
If the user presses the 'Analyse your mock test performance' button, a screen opens with two options. 'Previous test results' and 'Performance by topic'. When performance by topic is pressed, a very similar screen to the practice screen opens with the same list of topics and same information about each topic. However, the difference is that the information, used to populate this screen, is from the results of the mock tests rather than the practice questions.

I think that the second 'performance by topic' screen could be a little confusing because it's not immediately clear that it contains different data to the other 'performance by topic' screen. To improve this, in my system, I would have just one performance by topic screen and a drop down to change between the different data sources.

If the 'Previous Test Results' button is pressed the screen on the right opens. This screen shows the user all of the mock tests that they've attempted. They can easily see their score, how long it took them to complete the test, and whether they passed or failed the test. If they press the graph button, they can see a graph of all their previous results.

Back	Previous results	Graph
9 May 2018	47 out of 50 answered correctly - 8:20 mins	PASS
8 May 2018	46 out of 50 answered correctly - 7:46 mins	PASS
8 May 2018	48 out of 50 answered correctly - 5:42 mins	PASS
8 May 2018	48 out of 50 answered correctly - 5:52 mins	PASS
7 May 2018	49 out of 50 answered correctly - 6:13 mins	PASS
6 May 2018	46 out of 50 answered correctly - 5:03 mins	PASS

Study
 Practice
 Mock test
 Extras
 Settings



This makes it clear to them how they're progressing. It's also very useful that the pass boundary is marked on the graph so the user can easily see which tests they passed and which ones they failed.

Throughout the theory test section of the app, there is a bar of options at the bottom which are almost always displayed (shown below).

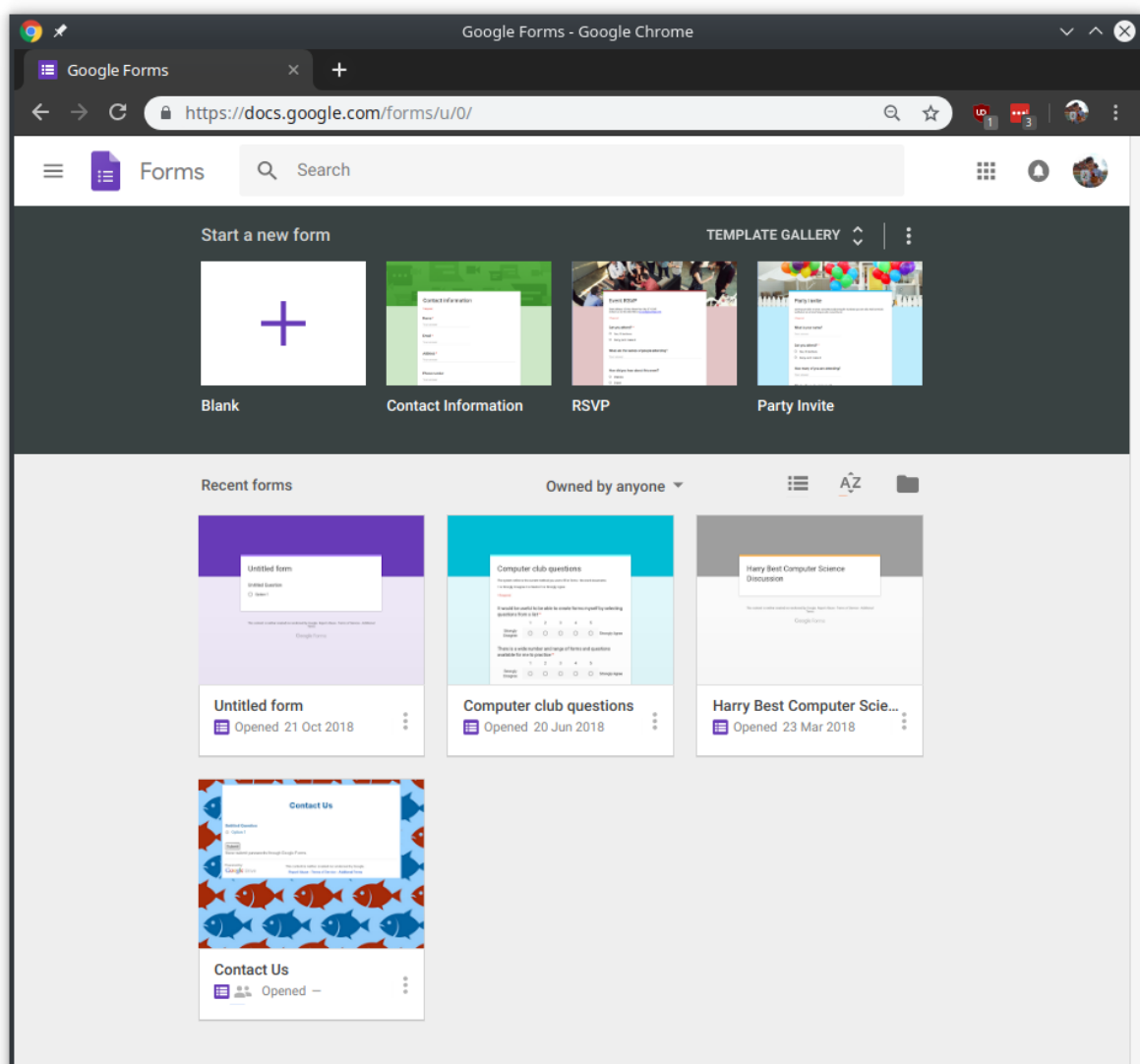


This makes it really easy for a user to navigate through the app because it allows them to quickly switch between different sections. Also, because they're almost always visible, they're usually on hand to the user so they can quickly be used. It also means that the user always knows where the buttons are. The only times that they're not visible are when the user is completing practice questions or is completing a mock test. The bottom bar also doesn't appear in the graph screen. This could be confusing for the user as they'd expect it to be there.

Google Forms



Google Forms is a google product that allows users to create and distribute online forms. The user can create a form by adding multiple different questions (and types of questions to it) and then can send it to others to be filled in online. The form creator can see the results of the form in real-time and graphs are automatically plotted to represent the data. I will look at just the form creation aspect of the system as that is the only element that is relevant to my system.



When the system is opened (i.e. you navigate to forms.google.com) a screenshot of each of the forms that the user has previously created is shown so that they can open them quickly again. I think that this a good feature because it means that the user can quickly edit a form and the visual preview means that they can tell exactly what the form was. However, I don't think that this feature would be useful for my system because all of the forms will look very similar so it would be difficult to tell them apart just based on screenshot. At the top there is also a feature to search through the forms that they've created. There is a distinct black strip at the top to differentiate the create new form buttons. I think that it's good that the buttons to create a new form are so distinct from the rest of the buttons this makes it really clear that they serve a different purpose. The buttons allow the user to create a new blank form or one from different premade templates. The inclusion of premade templates is also good and could decrease the time that it takes the user to make a form.

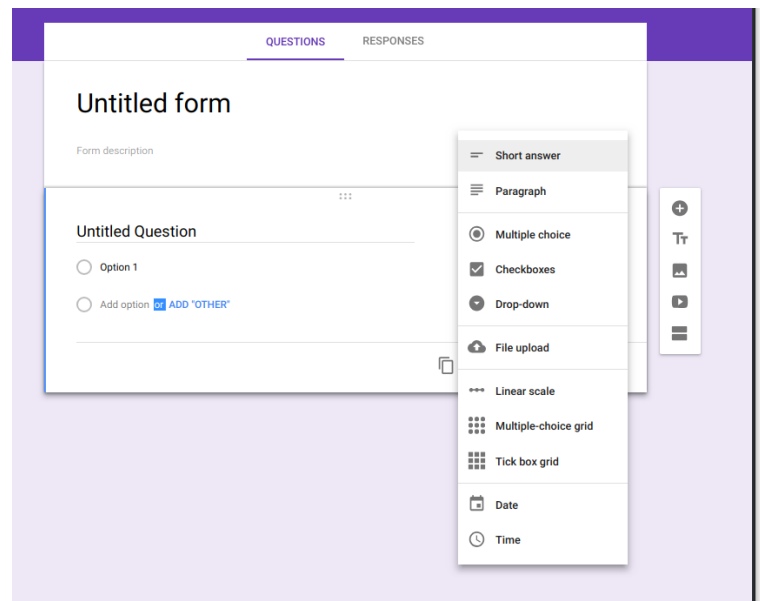
Once you press create blank form you are taken to a blank form to create. There is already a basic multiple choice question added. I like the large title that the form has because it makes the title of the form very obvious to the user. The fact that a question is also included makes the system easier to use because it makes it clear where questions of the form go.

I like that the title and the description of the form can be typed right into the form and don't have to be entered using dialogs because I think that it simplifies the process.

The required toggle in the bottom right of the question makes it very easy to mark a question as required. Ability to copy questions is good because it allows a large amount of similar questions to be quickly made.

I think that it's not immediately clear what the five buttons/icons on the right of the question do. However, they have descriptive and clear tool tips so they're still relatively easy to use.

When the add question button is added a new multiple choice question is added to the form the user must then select the type of question required using a dropdown on the question. I think that this is a complicated and confusing method of adding questions and that just having the system prompt for the type of question to be added when pressing the add button would be better.

The screenshot shows the Google Forms editor interface. At the top, there are tabs for 'QUESTIONS' and 'RESPONSES'. The main area is titled 'Untitled form' with a 'Form description' field below it. A question titled 'Untitled Question' is being added, showing 'Option 1' and an 'Add option' button with a blue plus icon and the text 'ADD "OTHER"'. On the right side of the question, a dropdown menu is open, displaying various question types with their corresponding icons: 'Short answer', 'Paragraph', 'Multiple choice' (selected), 'Checkboxes', 'Drop-down', 'File upload', 'Linear scale', 'Multiple-choice grid', 'Tick box grid', 'Date', and 'Time'. To the right of the dropdown is a vertical toolbar with icons for adding a new question, deleting a question, and other actions.

However, what I think is good about this menu is that, rather than just having to rely on the description of the question type, there is an icon that represents each question type. This makes it very clear what each type of question is even if the user isn't familiar with the names.

Although Google forms makes it easy to create questions I think that it would be beneficial if it had premade questions that could be added to forms. This would speed up the form creation process because common questions such as 'Address' can be quickly added without having to spend the time creating them.

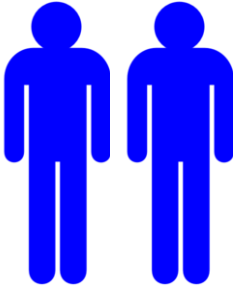
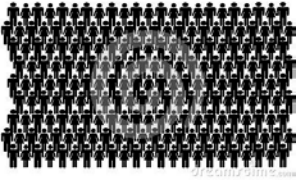
I like the simplicity of the drag and drop system of rearranging questions however I think that it can be difficult to use when the form is very long as the window also scrolls as you move the question this makes it hard to precisely and correctly place the question first time.

Best Features / Facilities

Feature/Facility to include	Source
Explanations of how to complete each type of question.	Digital Unite
Explanations of common errors in forms that pop up when the user makes one of those errors. E.g. if they press enter before the form is fully filled a dialog box pops up asking them if they really meant to press enter.	Digital Unite
Simple and clear breakdown of the user's past scores.	Official DVSA Theory Test Kit App
Consistent navigation options shown throughout the program in order to allow the user to easily navigate through the app	Official DVSA Theory Test Kit App
An easy to use menu to select practice questions with a clearly labelled start test / practice form button.	Official DVSA Theory Test Kit App
Ability to enter the title and description of the form without using dialog boxes.	Google Forms
Consistent method option entry between Radio Buttons, checkboxes, and dropdowns.	Google Forms

2.3 Proposed System

2.3.1 Stakeholder Requirements for New System

Group leader (Steph)	Description
	• Steph needs to be able to use the system to easily create realistic forms that the users can attempt. This should be done by selecting the practice questions from a list and the system should do the rest. • Steph should be able to easily create new practice questions that can be added to forms. • Steph should be able to export the forms and copy them to a USB stick so that she can easily distribute them to the instances of the system running on each user's computer. • Steph should be able to use the system to automatically produce reports on each user so that she can see how they're progressing. • Steph should have an admin account that lets her perform more complex tasks such as creating new questions. • Steph should be able to easily track the attendance of the users. • Steph should be able to store, access and modify the personal details of each user.
Volunteers	Description
	• Volunteers should be able to easily see what questions a user is struggling on so that they can effectively help them. This should be done by allowing the volunteer to see a report, produced by the system, showing what questions the user is struggling with. They won't need a unique login for this as they should be able to do it from the user's account.
Users	Description
	• The users should be able to easily use and navigate the system. • They should be able to practice realistic forms that are properly validated so that they understand where they've made errors and can improve. • They should be able to see a report of their weak areas and types of questions that they commonly get wrong so that they can do more focused practice on a particular area. ○ This report should also show their attendance. • They should be able to use the system to automatically create new practice forms for them. They should just select the questions that they want in the form from a list and the system should do the rest. • They should be able to use the system to automatically create practice forms that are based on their weaknesses. • They should have a unique login that they use to login to the system.

2.32 Specification for Proposed System

The purpose of the system is to help the elderly learn how to fill in online forms by providing a large number and wide range of forms that the user can practise. The system should also allow the creation of new questions and forms. This is to extend the range of forms that the users can attempt and increase the longevity of the system. Being able to fill in online forms is an increasingly essential skill as an increasing number of forms are being moved online.

In brief, the system should allow each user (the elderly member of the club) to login. After the user has logged in, the system should allow them to either create a form, attempt a form, or view statistics about their progress completing forms.

If the user chooses to create a form, they should be able to do this by selecting the questions that they want in the form from a list. The system then should automatically create and format the form so that they can attempt it. This means that the user will be able to easily create a large number of custom forms and so will expand the range of forms that they're able to practice.

If the user chooses to attempt a form, they should be able to select the form that they want to attempt from a list of forms in the system. The form should be fully validated so that it's as realistic as possible and so that the user is aware of when they're making mistakes. This is the main feature of the system and, through practising forms like this, the user should be able to improve their skill and confidence in filling in forms.

Finally, if the user chooses to view statistics about their progress completing forms, they should be shown various graphs and charts which show data about how they're progressing such as how long they spend on each question type and how many times they've failed the validation checks for each type of question. These graphs should also have any times that the user has been absent marked on therefore it can easily be seen if regions of low progress are because the user was absent.

Steph should have an admin account that she can login with. From this account, she should be able to produce a report on how a user is progressing (exporting the data from the view statistics screen). She should also be able to create new forms.

A feature that only Steph will have and not the users is the ability to create new questions. Steph should be able to select what elements she wants in a question such as a text field or radio buttons and then save the question and add it to forms.

Steph will also be able to create new user accounts and store information about each user. Steph should be able to log in to any instance of the system and create new user accounts for that instance.

When Steph logs in, there should be a button to create a new user account. She would then be asked for the username and password of the new user along with their first and last names and their date of birth.

On her instance of the system, Steph should be able to store information about each user. Steph should be shown a list of all users (this would have to be manually inputted - the user accounts wouldn't actually exist on her system and would only exist on the user's instance of the system) and then would be able to select a user to view information about them. The information stored should be their first and last names, their attendance, their phone number, and date of birth. Steph should also be able to amend any details by clicking an edit button. She would then be able to press a save button to write the changes to file.

To track progress, Steph should be able to press a take register button from her main menu and then a list of the users would appear allowing her to mark with a check which ones were present.

The user also needs their attendance tracked on their instance of the system. To avoid having to copy the database from Steph's version of the system to the user's instance, the system should track automatically when they log in. When the user logs in, the system should store to a file that the user has logged in along with the date. If a user logs in multiple times in one day, the system should discard these extra logins. This means that the system can automatically track a user's attendance. If the user isn't present for a week, when they log into the system, the system should ask, using a dialog box, if they missed the last session or if the club wasn't running. This means that the system will be able to determine that the user has missed a session because the club wasn't running and discard this from their absences.

As there will be multiple instances of the system running on different computers, the forms and questions from the system should be able to be easily exported and stored on a USB stick and then imported into the different instances of the system to transfer forms between them. Single forms and questions should also be able to be exported and imported on the system.

The users should be able search and sort the lists of forms and questions in the system so that they can more quickly and easily find the forms and questions that they want to practice.

The interface of the system needs to be simple and easy to use. This is because the users have very limited computer skills. This will be achieved by having clearly labelled buttons and menus and there should be very few nested options.

The computers that are used at the computer club are slow and old laptops. This means that it's essential for the system to be well optimised as it will still have to run well and be responsive. The computer club only runs for 1 hour each week therefore, if the system is slow, a large amount of time will be wasted just waiting for it to process actions. The user should never have to wait more than 20 seconds for an action to be completed. The majority of actions such as searches and sorts should be completed within 5 seconds. The more intensive actions such as loading and saving the databases are allowed more time (around 15-20 seconds) as they are only going to be run once during the session - loading would be run at the very start and saving at the end of session.

2.33 Methods to be used in Proposed System

Method	Technical Justification
Swing GUI Library	There is a library called Swing that has been developed for Java which allows the easy creation of GUIs. It contains several standard components such as text fields and dropdowns that can easily be used. This will mean that I can quickly and easily create a GUI for the system because I just have to put premade elements together rather than creating them from scratch. Because they're objects, I can easily create multiple graphical components that I need for the system by instantiating different objects. Also, all of the technical implementation is hidden within the objects (encapsulation) therefore I am able to use them with little/no knowledge of how they work. This saves time and makes the overall process of creating a GUI much simpler. It's important for the system to have a GUI because the users are not very computer literate and would be confused by and unable to use a command line interface. If the system didn't have a GUI, it also wouldn't be able to correctly and accurately mimic online forms.
Data structures	I am going to use and build upon the basic data structures that Java provides such as Arrays. Java allows the creation of an array for any type of object (but only a single type per array). This means that I can effectively organise groups of objects. It will allow multiple questions to be easily associated with each form. It also allows the storage lists of forms and questions that can be searched through and sorted. In addition I won't need individual variables for each different form / question which makes the code less complex and easier to read and maintain.
File handling	The Java.IO library provides several ready-made classes that can be used to handle files. I will be making use of this in order to save the forms, questions and user data permanently to the hard drive. As these are quite complex tasks, it's useful that these are premade classes that I can easily implement as it will speed up development time and as they're already thoroughly tested it will mean that there are fewer bugs. Saving files to the hard drive means that progress can be saved between closing and opening the system. It also means that the forms and questions that are part of the system don't have to be hard coded as they can just be loaded from the file. This means that the code of the system won't need to be altered to add / change questions or forms.
Local and Global Variables	Java allows the use of local and global variables. The use of global variables means that I can make certain variables accessible to the entire program/module. This is useful for GUI elements, for example, as they will need to be referenced at multiple stages of execution and from many different parts of the program. Local variables mean that I can have a more readable and maintainable code base. Local variables only exist for the particular subroutine that they're used therefore I can use the same identifier for many different local variables. This is advantageous for common variables such as 'searchTerm' as I won't need to name them searchTerm01, searchTerm02 etc. as they'll be

	used in different methods. This will avoid cluttering the code with multiple variables that have very similar names and are used for the exact same purpose.
User Defined Methods and Classes	Java allows the creation of custom methods and classes. I will be able to define custom methods that can be used at many different stages of execution. This will be useful for actions that will need to be repeated multiple times such as adding a question to the database. It will also aid the readability and maintainability of the system because it means that there will be minimal redundancy in the code. Also if I find a bug or issue with a particular method I will only need to fix it once rather than having to look through the entire code and fix each section of code individually. I will create many different classes and each class will contain the subroutines/methods that are relevant to it as well as any data. This makes it very easy to process and handle objects because each object will already contain the processing required and any data. Therefore, I won't need to write processing methods in many different parts of the program. Even if the object is passed between different parts of the program, the processing and data that is associated with it won't get lost and so there will be no loss of integrity and the object will be able to be processed and handled just as easily in that part of the program.
Validation	Validation is an essential part of my system as the entire system is based around teaching the users how to fill in forms. Java provides many premade and easy to use methods to compare data. For example, I will be able to use the <code>.equals()</code> method to compare two strings to see if they're equal and the <code>.length()</code> method to easily get the length of the string. I can also use these methods in conjunction with some arrays of valid options in order to easily perform lookup checks. The test string to be validated would be compared with each item in the array in turn and if it matches any of the strings in the array the validation check will pass.
The Java Virtual Machine	Java programs run in the Java virtual machine, rather than directly on the hardware, and this means that the system will be able to run on multiple different operating systems without any alteration to the code. If the computer club moves away from Windows to Linux or MacOS the system will still work. This extends the lifetime of the system. It also means that I won't have to deal with or account for differences between OS or CPU architectures. This will save development time as the system will be less complex to develop as these will be handled and managed automatically for me. I also currently don't have the required programming skill to account for different architectures.

2.34 Objectives

#	Objective	Success Criteria	Required Performance
	Questions		
1	Ability to create Questions	The admin should be able to create new questions by dragging and dropping elements such as text fields and radio buttons from a toolbar menu on to a blank panel.	The dragging action should be smooth and the question should be added to the system and saved to the question database file within 5 seconds of pressing the save button.
2	Ability to add labels	Labels should be able to be added to the question so that the fields and other components can be labelled. E.g. "Username:" or "Email Address:" Once the user selects add a text field a window should appear asking what the text of the label should be.	The window asking the user what the text of the label should be should open within 2 seconds. The label should be added to the form within 2 seconds.
3	Ability to add text fields	Text fields should be able to be added to the question.	The text field should be added to the question within 2 seconds.
4	Ability to specify the type of data that should be entered into the text field	The type of data that the text field should accept should be able to be specified through a dropdown menu presented to the user. This is so that the text field can be correctly validated, e.g. labelling a text field for email entry so that the text field has a format check applied to check that an email address has been entered.	The menu to select the type of validation should open within 2 seconds of adding a text field.
5	Ability to add radio buttons	Radio buttons should be able to be added to questions. There should be a prompt that appears asking the user to enter the options that they want to be able to be selected by using the radio buttons.	The window prompting the user to enter the options should open within 2 seconds and the radio buttons should be added to the form within 2 seconds of entering all of the options.
6	Ability to add drop-downs	Drop down fields/menus should be able to be added to questions. There should be a prompt that appears, asking the user to enter the options that they want the dropdown to have.	The window prompting the user to enter the options should open within 2 seconds and the dropdown should be added to the form within 2 seconds of entering all of the options.
7	Ability to add checkboxes	Check boxes should be able to be added to questions. There should be a prompt that appears, asking the user to enter the options that they want to be able to be selected by using the checkboxes.	The window prompting the user to enter the options should open within 2 seconds and the checkboxes should be added to the form within 2 seconds of entering all of the options.
8	Ability to add password fields	Password fields should be able to be added to the question.	The password field should be added to the question within 2 seconds.
9	Ability to add	A calendar that can be used to select a date	The calendar should be

	calendar entry	should be able to be added to the system.	added to the question within 2 seconds.
10	Ability to add a document upload field	There should be a field that allows documents to be uploaded to the form. This would be mostly used when the user has to upload a picture to a form however other types of files such as pdfs should also be able to be uploaded.	The document upload field should be added to the form within 2 seconds.
11	Ability to delete Questions	The admin should be able to delete questions from the database by selecting them and pressing a clearly labelled delete button.	The question should be deleted from the database within 2 seconds of pressing the delete button.
12	Ability to sort Questions by difficulty	The question table should be able to be sorted, via a bubble sort, by the difficulty of the question, either ascending or descending to allow the user to easily locate questions. This should be done by pressing a clearly labelled button.	The sort should be completed and the results displayed to the user within 2 seconds.
13	Ability to sort Questions by type	The questions should be able to be sorted, via a bubble sort, by type alphabetically. This should be done by pressing a clearly labelled button.	The sort should be completed and the results displayed to the user within 3 seconds.
14	Ability to filter Questions by difficulty	A linear search should be employed to allow the user to search for questions of a particular difficulty. The user should be able to select the difficulty that they want via a slider.	The filter should be completed and the results displayed to the user within 2 seconds.
15	Ability to filter Questions by type	A linear search should be employed to allow the user to search for questions of a particular type. The user should be able to enter the search terms in a text field.	The filter should be completed and the results displayed to the user within 2 seconds.
16	Email addresses, dates, and phone numbers should have format checks	A format check should be applied to fields which request email addresses, dates or phone numbers, to ensure that the user has entered valid input. If the data fails the check, the user should be alerted via a red outline / mark next to the questions that they failed. The user should be able to amend their answers to the form and resubmit.	This check should be performed and the results displayed to the user almost instantly after the user presses the submit button.
17	Addresses should be looked up via the google maps API to autofill information	The street number and postcode that the user has entered should be queried with the Google Maps API so that the user can select their address from a drop-down as is common with online forms. If the address can't be found, the user should be alerted and should be given the option to manually enter an address.	This lookup should take no more than 10 seconds on an average internet connection. The menu for the user to select an option should also be displayed within these 10 seconds.
18	Passwords should have double entry verification	The passwords that the user enters should be verified via double entry verification. If the passwords don't match, the user should be alerted.	This check should be performed and the results displayed to the user almost instantly after the user presses the submit button.
19	All fields should have a presence check	All of the fields in a form should be presence checked to ensure that the user has entered an option. This includes checking that the user	Each field should take less than a second to check.

		has ticked checkboxes and radio buttons, and has selected an option from the drop down components/combo boxes.	
20	Document upload fields should only allow certain types of files to be uploaded	When creating a document upload field, the admin creating the question should be able to specify which type of file e.g. text document or picture should be allowed to be uploaded. The admin won't have to enter the specific file extensions that are allowed and can simply pick either picture or text document, for example. The field should be validated and not allow users to upload files of the wrong type.	The popup to specify the type of files allowed should appear within 2 seconds of adding the field to a question. The validation check should be completed within 2 seconds.
21	Ability to export a Question to file	A particular question should be able to be exported to a file in order to be imported in another instance of the system. The file should be able to be saved to a USB stick and transferred between computers.	The export process should complete and the question should be saved within 5 seconds
22	Ability to import a Question from file	A question that was previously exported should be able to be imported into a different version of the system. This allows transfer of questions between instances of the system.	The question should be imported and added to the database within 5 seconds
23	Ability to attempt a question	If the user wants to attempt a question the question should be displayed to them.	The question should be opened within 2 seconds
24	Ability to submit a question	If the user wants to submit the question it should be validated appropriately. If it fails the check this should be highlighted to the user.	The validation checks should be completed within 2 seconds
	Forms		
25	Ability to create forms	The user/admin should be able to create forms by specifying which questions should be in them. All the formatting and layout should be automatically completed by the system.	The form should be created within 5 seconds of the user pressing save form after selecting the questions that they want in the form.
26	Ability to create forms from user weaknesses	The user should be able to press a button which will automatically add the questions that they struggle with (those which take them the greatest number of attempts to correct their errors after getting the question wrong) to a form. All the formatting and layout should be automatically completed by the system.	This form should be created within 10 seconds of the user pressing the make form button.
27	Ability to sort forms by difficulty	The user should be able to sort the forms by difficulty via a bubble sort by pressing a clearly labelled button.	The sort should be completed and the results displayed to the user within 2 seconds.
28	Ability to filter forms by difficulty	A linear search should be employed to allow the user to search for forms of a particular difficulty. The user should be able to select the difficulty that they want via a slider.	The filter should be completed and the results displayed to the user within 2 seconds.
29	Ability to filter forms by types of questions tested	A linear search should be employed to allow the user to filter for forms which contain particular types of questions.	The filter should be completed and the results displayed to the user within

		The user should be able to select the types from a drop-down. These skills would correspond to the types of the questions in the form.	2 seconds.
30	Ability to edit forms	The admin should be able to alter the forms that are currently part of the system. They should be able to change the difficulty rating, and the questions that are in the form.	The changes to the form should be immediately applied and saved to the database.
31	Ability to delete forms	The admin should be able to delete forms from the database by selecting them and pressing a clearly labelled delete button.	The form should be deleted from the database within 2 seconds of pressing the delete button.
32	Ability to export forms to file	A particular form should be able to be exported to a file in order to be imported in another instance of the system. The file should be able to be saved to a USB stick and transferred between computers.	The export process should complete and the form should be saved within 5 seconds
33	Ability to import forms from file	A form that was previously exported should be able to be imported into a different instance of the system. This allows transfer of forms between instances of the system.	The form should be imported and added to the database within 5 seconds
34	Ability to attempt a form	If the user wants to attempt a form, all of the questions related to that form should be opened and displayed to the user, allowing to user to attempt them.	The form should be opened within 5 seconds
35	Ability to submit a form	If the user wants to submit the form, all of the validation checks of each question should be triggered. If they all pass, the form is complete and submitted. Otherwise, the user should not be able to submit the form and the questions that were incorrectly filled in should be highlighted.	This check should less than 10 seconds
	User		
36	The time taken to complete each question should be stored	All of the past times that it has taken the user to attempt a particular question should be stored. These should be saved in a CSV file. This progress tracking and saving should be performed automatically.	The saving and loading operations should be completed within 2 seconds.
37	The number of times that they've failed the validation check for a question should be stored	The number of times that the user has failed a validation check for each question should be stored in a CSV text file. The number of times that it has taken the user to correct an error (that caused them to fail a validation check) in their response to a particular question on a form should also be stored. This progress tracking and saving should be done automatically.	The saving and loading operations should be completed within 2 seconds.
38	Ability to add users	The admin should be able to add new user accounts to the system. Each user should have a username and password	The new user should be saved to the database and added to the system within 2 seconds.
39	Ability to edit users	The admin should be able to edit a user's username, password and any other information that is stored about them.	The updates should be saved and applied within 2 seconds.

40	Ability to delete users	The admin should be able to remove user accounts from the system.	The user should be removed from all databases and the changes should be saved within 5 seconds.
41	Ability to search through users	A linear search should be employed to allow the admin to search for users by name. The admin should be able to enter the search terms in a text field.	The search should be completed and the results of the search should be displayed within 5 seconds of the admin pressing the search button.
42	Ability to produce a report on the user's weak areas/progress	A report should be able to be produced which details the weaknesses of each user. The number of times that they failed the validation check for each question type, the average time that it takes them to complete each question type, and the number of times it's taken them to correct an error on average for each question type should be outputted. This should be produced when a user clicks a clearly labelled button and the admin should be able to produce reports for multiple users at once. Reports should be able to be printed.	This report should be produced and outputted within 10 seconds.
43	How much of each form a user has attempted should be tracked.	While a user is completing a form, the system should keep track of what percentage of questions they've completed. Once the user closes the form or completes the form, this data should be stored. This is so that the user can easily see what forms they've already completed and which ones they have left to do.	The data should be stored to the database within 5 seconds of the user completing the form or closing the form.
44	Firstname, lastname, Date of birth and phone number should be stored for each user	This data should be stored on the admin's instance of the system and should be encrypted and only accessible to their account. This database should be automatically loaded and decrypted when they pressed the view users button.	The database should be loaded and decrypted within 10 seconds. The database should be encrypted and saved within 10 seconds.
45	The attendance for each user should be automatically tracked	When the user logs in, the system should store the date to the attendance file. If the user logs in multiple times in one day, the date should only be stored once in the file.	The date should be added to the attendance file and the changes saved within 10 seconds of the user logging in.
46	The system should determine the reason for a user missing a session	The system should determine if a user has missed a session because the club wasn't running. This should be done by a dialog box that pops up when the user logs into the system after they have been absent for a week.	This box should open within a 3 seconds of the user logging in.
	Interface		
47	Should be simple	The interface should be easy to use and simple. There should be few nested menus and all the actions that the user would want	The user should be able to find and access an option/menu/screen within

		to take should be clearly presented.	a 2 minutes if it's their first time using the system.
48	Login page which allows users or admins to login	There should be a login page which allows the admin and users to log in to the system. Their usernames and passwords should be checked against stored credentials in the database. The user/admin should not be allowed access to the system if their credentials are invalid.	If the user has provided correct credentials, the logon process should be completed within 5 seconds and the user should be presented with the main menu.
49	There should be a main menu panel	The main menu panel should provide buttons for the user to press which takes them to different areas of the program.	Each panel available to open from the main menu window should open within 5 seconds.
50	There should be a panel which will display the forms to the user and allow them to search through them	The user should be able to press a button in the main menu panel which will open a panel that displays the forms to the user in a table. The user should be able to sort the table by pressing a button and should be able to search for a form by entering their search terms into a text field and pressing a clearly labelled search button.	The table should be updated with the results of the sort/search within 2 seconds.
51	Allow user to select a form to be attempted	The user should be able to select a form from the table in the view forms panel and press a clearly labelled button to open it and attempt it.	The window containing the form for the user to practice should open within 2 seconds.
52	Allow form to be submitted	There should be a submit button at the bottom / end of each form. When the user presses the button the submit form process should be invoked.	The submit process should be invoked within 1 second.
53	There should be a panel which will display the questions to the user and allow them to search through them	The user should be able to press a button in the main panel which will open a panel that displays the questions to the user in a table. The user should be able to sort the table by pressing a button and should be able to search for a question by entering their search terms into a text field and pressing a clearly labelled search button.	The table should be updated with the results of the sort/search within 2 seconds.
54	Allow user to select a question to be attempted	The user should be able to select a question from the table in the view question panel and press a clearly labelled button to open it and attempt it.	The window containing the question for the user to practice should open within 2 seconds.
55	Allow question to be submitted	There should be a submit button at the bottom / end of the question. When the user	The validation process should be invoked within 1

		presses the button, the question should be validated.	seconds.
56	Questions that have failed their validation check should be highlighted	If a question fails its validation check, elements of the question should turn red / have a red outline to show that it's been filled in wrong.	A question should change and update its state and appearance within 2 seconds.
57	The user should be able to create forms by selecting questions from a list	The user should be able to select questions from the view questions panel by pressing a button to add them to a list of questions. From this list, a form should be automatically created.	After selecting the questions and pressing the create form button, the form should be created within 10 seconds.
58	There should be a statistics screen which shows the user's progress	There should be a window which displays the user's progress to them and other statistics about how they are performing, such as the number of times that they've failed the validation checks and the average time to complete each question. The time that it's taken the user to complete a question compared to previous attempts should be plotted with a line chart. The percentage of times that a user has failed a validation check for a question should be displayed as a pie chart.	The window should open and all the graphs should be plotted and displayed within 15 seconds.
59	There should be a context sensitive help button available in each window and panel of the application accessible by a regular user	The user should be able to press a help button which has the icon of a question mark to open a window which describes how to use the screen/section of the system that they currently have open. The button should be in a consistent place in every window, and panel, and should have the same icon and size so that the user can always easily locate it. There should not be help available in the admin only areas of the application as they will not require it.	The help menu should open within 2 second of the button being pressed.
60	The admin user should have a different main menu	The main menu that the system opens up to should have a few extra options for the admin user. The admin should have: ● A button which allows them to open a window to view information about each user. ● A button to create questions. ● A button to take a register	These changes should be applied almost instantaneously when the admin user logs into the system. The windows/panels that correspond to the buttons should open within 5 seconds of the button being pressed.
61	The admin user should be able to view information about each user	There should a be panel which displays a list of all the users to the admin user. They should then be able to select an individual user or search through the list to find the user that they want. Once a user has been selected the panel should show them all the data that's stored about that user. This would be their names,	The panel should open within 4 seconds. Once the user has been selected, the information about that user should be displayed within 5 seconds.

		phone number, and date of birth.	
62	The admin user should be able to edit information about each user	In the view information about a user screen, there should be an edit button so that the admin user can amend any details if necessary. The labels/text should change to text fields so that the admin can change the data. There should be a save button so that the admin can save the changes.	Once the save button is pressed, the changes should be saved and applied within 5 seconds.
63	The admin user should be able to take a register	There should be a window that displays all the users to the admin in a list. Each user should have a checkbox next to their name that the admin can tick if the user is present.	The register should open within 4 seconds, and the changes should be saved within 3 seconds.

Appendix

Phone Interview

Harry: How does the user currently find and attempt a form on the system?

Steph: They have to go through to public services and then look at all the forms that are in the public services section.

Harry: So is that on learnmyway?

Steph: That's on learnmyway yes, that's the only one where I know how to access any forms like that.

Harry: What about if it's just a word document that you've made.

Steph: A word document that I've made will generally will be on my pendrive.

Harry: And then do you just copy it to the user's desktop and do they open it like that?

Steph: Yeah or to their pendrive.

Harry: So would you say that it's a realistic way of doing it or could it be improved? Does it correctly mimic the real world?

Steph: I think it's very similar looking through a website and looking through a file explorer as they still have to look through different forms.

Harry: So how do the users know that they've completed the forms correctly?

Steph: Well on that one there's only two pieces that they can complete and if they've not completed it correctly they can't move on.

Harry: So that's on learnmyway so what about the word document?

Steph: On the word documents there's no way of knowing if it's done correctly. I just have to have a look at it and see and tell them if they've done something wrong. Neither of them really mimic the real world.

Harry: So you'd say that's an area that needs to be improved?

Steph: It is definitely yeah.

Harry: What do you like about the current system, learnmyway and your word documents?

Steph: I like how it's very user friendly, it's got audio and it's also got if you can't listen to it it's got the words for it. It's very plain speaking that's what I like about it. It doesn't presume that you know something or anything like that it's for the very beginners.

Harry: What do you think the main problems are with the current system?

Steph: The main problems are when they're completing forms they're not starting the forms at the beginning working their way through it and submitting it. All they're doing is they're put on say the blue badge form it already comes on as if part of it's been filled in. You don't have to create an account or anything like that and there's no submitting because you only have to answer maybe one or two questions.

Harry: So are these the main things that you're looking for with the new system?

Steph: That's the main things we're looking for. Whether we're looking for the form through file explorer or through a website it doesn't matter. But it's actually opening up the form from the off filling in the questions, do you know the different types of questions that you've got there might be a dropdown category there might be a calendar category that you click on and a calendar comes up you know lots of different things like that. And then also you've completed if you've had to upload a photograph you upload the photograph and then it's submitting it. I mean it may not go anywhere it doesn't matter they've actually completed it and submitted it. And then if they've not filled in anything right it won't let you submit it and it'll go back to something that's not correct. That would happen in the real world.

Harry: What data do you store about the current users so each member? Maybe you might store the attendance or the progress.

Steph: I always do a register. Phone numbers, things like that.

Harry: Is there anything you store as part of the form filling in that isn't just the general club

Steph: I don't, I don't have any kind of capabilities for that. I do have an access database but I don't use it to the fullest. But it's very difficult to start saying I do every week or I do that every week or this person has done this.

Harry: So do you think it'd be useful for the system to track progress automatically? Like what forms they've done and what questions they've done.

Steph: I would find it very useful if there was something of a record of what everyone's done. Yes, that'd be brilliant. That would be good for them as well.

Harry: So do you think it'd be useful for the system to give you a small tutorial on how to fill in each type of question?

Steph: Very small not a lot. They tend to learn better with just by doing it. If you think of the group a lot of them have early onset dementia. So when you actually are talking to them and going through a tutorial they're not really taking it in. It's better to have a little bit of a written thing for them straight forward and then they do it and usually have someone sat with them helping them. So maybe a little bit of tutorial but not much. They're not going to take in that amount of information.

Harry: Is the ability of the system to allow you to create new forms by selecting the questions that you want in the form useful? So what I mean by this is there would be a list of all the question types and you can just tick the ones you want and then you click create form and it makes it all into a practice form for you.

Steph: That sounds very simple. That would be brilliant.

Harry: What progress would you like tracked about each user? Like time taken to complete a question and the number of times that they've failed a validation check. Would you like anymore?

Steph: No I think that'd be it. Maybe a progress chart of how long it has taken them to fill it in perhaps something that if they've done anything wrong how many times it's taken them to correct it. Because if they do something wrong and they've not been able to submit it the next time they go wrong have they learned from it? They may not tell you that but you might be able to compare the two and think well yeah they did that last time and they've not done it this time.

Harry: So I mentioned about creating forms before by selecting questions and ticking the ones you want. So along with this progress tracking what do you think about just one button that would automatically create a form for a user which has all the questions that they commonly get wrong in so they can just practice them all right there?

Steph: Yes. That would be good

Harry: The last question is what other improvements would you like with the new system if I've not mentioned something?

Steph: I don't think there is.

Text interview

Harry: What do you currently store generally about each user? I know you store attendance and phone numbers what else is there of that sort of thing

Steph: Yes, dates of birth and email addresses and passwords

Harry: What do you do with the data and why do you store it?

Steph: We store the dates of birth etc. confidentially, and the register, sometimes we don't see people for a while and we can check the records of all the groups and see if they've attended or not. Email and passwords are stored because you can guarantee some people will forget them. All personal info is stored safe and confidentially

Harry: Are the emails and password the emails and passwords to the learnmyway system?

Steph: Yes, and their own email addresses. I only store them if I think they may forget them

Harry: Are all these pieces of information stored on paper or on the access database?

Steph: I store them on my machine on the access database

Harry: I noticed when I was observing that you had a ring binder of paper attendance logs and you copied them from the paper to the access database are there any other manual processes you do like that of searching/sorting/copying data?

Steph: Yes, names and numbers of new clients are stored in the ring binder, I then move them over to the database

Harry: So when you get a new client (is this a user/elderly member?) You give them a paper form to fill in then you store it in the binder and copy their details to the access database?

Steph: No I just take their name and number. If they become an official HBS Service User Jackie & Carol take their confidential information and store them in a locked filing cabinet

Harry: Ok how do you take their name and number is it verbally and is this the same for the email address and password?

Steph: Yes

Harry: When the user wants their email address and password how do you search through the database to find it.

Steph: I always have my laptop with me and I search on there by name

Harry: Are there any other sorts and searches you do in the database? E.g. do you ever sort alphabetically by name or filter by who attended the most recent session

Steph: I do both those searches

Observation

not 6 months working offline as no internet connection to current building.

0822: Arrived 10 computers. Set up all windows laptops. Only 2 volunteers Steph ~~knows~~ ~~knows~~

0829: Make tea + coffee for themselves

0833: Jackie (legid staff/coordinator) arrives

0830-0900

0840: Steph searches through ring binder of paper registers and copies into access database on her laptop. The ~~volunteers~~ volunteers talk between themselves.

0851: Another volunteer arrives.

0900: Volunteer arrives ~~Steph's~~ ~~Jackie's~~ man.

0908: Steph gets some USB sticks out.

0917: Steph starts to attempt to print some forms because there's no internet access.

0932: ~~the~~ 2 users arrive. One with their own laptop.

0934: Steph searches through a large binder of printed out lessons she's created.

0936: ~~the~~ User arrives

0938: user arrives

0936-0939: Steph copies a modup the badge form she's made on word with interactive elements to a USB stick to copy to the computers in the room.

0940-0949: Copying documents to laptops

0944: user arrives

0950: User arrives, and another volunteer arrives

0952: ~~the~~ User arrives
Volunteer arrives

Form is confusing as users have to find the sumers relevant to them rather than it taking them through properly.

Word document has no validation

0954: volunteer arrives

1000: lesson starts. 2 users in total.

1004: Tadhg leaves as she's busy and off another volunteer.

1004: Instructs users to find previously copied file on desktop and open it (word document)

1006: Steph explains that in the real world they'd have to login/register and click the start now button.

1007: Users struggle to find the correct page of the form as they have to gradually scroll down and it's easy to accidentally scroll past. Some users are confused about not clicking start now.

1008: Some users are struggling as some elements of the form should be check boxes but they're actually radio buttons.

1013

1013: Volunteer arrives.

1014: User with poor eyesight has to have a volunteer read everything out to them as the word document text is small.

with assistance from constant volunteer time. I use read all the

1017: Steph has to help a user. Element of form broken. Can't select yes -

1015-1014: User struggles to enter postcode. Keeps waiting for address selection to pop up like it does in the real thing confused when it doesn't.

1020: User misses some required elements.

1020: User struggles to enter data of birth as no year selection.

1021: Volunteer steps mum leaves

1022: Mockup confirm national insurance number text fields have no double entry & ~~validation~~ verification.

1023: Another user confused about address not automatically appearing.

1023: User doesn't understand ~~understand~~ understand 00/MM/YY date of birth entry

1029: Steph packs away some excess laptops as not everyone arrived / small group

1030:

~~1031: Check boxes on the form don't actually work as they're just square shapes.~~

1033: User finished the form but no submit button or validation.

1036: User who finished the form is just sat around with no further work to do.

1038: Steph helps user with broken checkbox in the form.

1038: another user has a problems with check boxes.

1039: Check boxes aren't actually check boxes they're just squares.

1040: Steph goes through saving a word document (the form) as most users have filled the form in as much as they can.

1041: Steph distributes each user ~~their~~ their own pendrive with their names on to save the forms to.

1047: Word has crashed on a user's laptop ~~so they~~ as the form is a little too much for word on the lowend laptop.

1047: User ~~struggles~~ struggles to reopen form.

1051: Another user ~~struggles~~ struggles to save the form.

1052: Most users have finished as much as they can with the form and are talking between themselves.

1055: Steph helps user with crashed word reopen the form. But it's a blank form and they've ~~lost~~ lost their progress.

1057: Steph ~~asks~~ asks users if they use online ~~banking~~ banking

1058: The users ~~turn~~ ^{begin to} off the computers as the session is almost over.

1100: packing up begins.

1103: Users put coats on and begin leaving

1103: ~~2~~ ⁴ ~~users~~ users leave

1104: ~~#~~ ~~was~~ ~~have~~ packing up

1105: user leaves.

1107: Some users + volunteers leave.

1109: Packing ^{up} finished

1110: Most people have left.

1115: I leave.

Achievements table

	id	achievement_name	Click to Add
+	1	Up to 4 Year Service	
+	2	5 - 9 Year Service	
+	3	10 - 19 Year Service	
+	5	20 Year Service	
+	6	Occasional	
+	7	Computer Group Certificate	
+	8	Computer Volunteer Thank you	
*	(New)		

A section of the Attendees table with sensitive information blurred

	id	Attendee	Address	Phone Num	Mobile Num	Email Address	password
+	78	Aida					
+	76	Alan Carter					
+	54	Andrew					
+	75	Anne Fare					
+	81	Audrey					
+	1	Barbara Tranter					
+	23	Bernie Crossland					
+	71	Betty Parkinson					
+	2	Bill Kenny					
+	45	Billy Whistance					
+	39	Brenda Bookthorpe					
+	62	Brian Parrott					
+	10	Carol Fildes					
+	93	Cham					
+	86	Chantelle (Chan) Gr					
+	88	Chloe Roberts					
+	3	Chris Demullet					
+	9	Chris McCoy					
+	84	Danny McKeown					

Course Names table

	id	Course Name
+	1	Starting Online
+	2	Online Basics
+	3	Online Plus
+	4	Workshop
+	5	Drag and Drop
+	6	Typing
+	7	Digital Learn
+	8	Asisting on a Computer

A section of the Meeting Attendees tables

id	meetingid	attendeid
1	14/07/2015	Chris Demullet
2	14/07/2015	Ralph Brandon
3	15/01/2015	Barbara Tranter
4	15/01/2015	Bill Kenny
5	15/01/2015	Chris Demullet
6	15/01/2015	Pat Kay
7	15/01/2015	Ralph Brandon
8	15/01/2015	Ron Mason
9	05/02/2015	Barbara Tranter
10	05/02/2015	Bill Kenny
11	05/02/2015	Ron Mason
12	05/02/2015	Jim Beagin

A section of the Meetings table

id	meetingdate	status	Week Number	Volunteer A	Friends	Subs Raised	Tea/Coffee	Costs	Sales	Amount Rai	Notes
87	28/04/2016	Active									
88	19/05/2016	Active									
89	16/06/2016	Active									
90	23/06/2016	Active									
91	30/06/2016	Active									
92	14/07/2016	Active			0						
93	21/07/2016	Active			0						
94	28/07/2016	Active	0		0						
95	11/08/2016	Active	0		0						
96	18/08/2016	Active	0		0						
97	25/08/2016	Active	0		0						
98	15/09/2016	Active	0		0						
99	22/09/2016	Active	0		0						
100	06/10/2016	Active	0		0						
101	13/10/2016	Active	0		0						
102	20/10/2016	Active	0		0						
103	10/11/2016	Active	0		0						
104	17/11/2016	Active	0		0						
105	24/11/2016	Active	1	7	14	£14.00	£4.50	£1.00	£23.00	£40.50	
106	01/12/2016	Active	2	6	17	£17.00	£3.75	£0.00	£0.00	£20.75	
107	08/12/2016	Xmas Party	3	14	15	£0.00	£0.00	£0.00	£0.00	£0.00	
108	15/12/2016	Closed	0		0	£0.00	£0.00	£0.00	£0.00	£0.00	
109	22/12/2016	Closed	0		0	£0.00	£0.00	£0.00	£0.00	£0.00	
110	29/12/2016	Closed	0		0	£0.00	£0.00	£0.00	£0.00	£0.00	
111	05/01/2017	Active	4	7	13	£13.00	£5.70	£117.64	£400.00	£301.06	Calendar etc. figures
112	12/01/2017	Active	5	5	8	£8.00	£4.00	£10.28	£24.00	£25.72	purchase for promotion
113	19/01/2017	Active	1	5	9	£10.00	£3.00	£100.00	£0.00	-£87.00	Rent Paid

With the plus button pressed

105	24/11/2016 Active		1	7	14	£14.00	£4.50	£1.00	£23.00	£40.50
id	attendeid	Click to Add								
913	Alan Carter									
914	Andrew									
915	Bill Kenny									
916	Brenda Bookthorpe									
917	Brian Parrott									
918	David Bellis - Seetec									
919	Dorothy Lawson									
920	Dot Riding									
921	Geoffrey Greenhalgh									

Training table

id	Attendee	Volunteer	Course	Date Start	Date Finish	LMW User	LMW Passw	Email	Email Password
*	(New)								

A section of the Volunteer Achievements table

id	volunteer	achievement	Date Achieved	Management
1	Afrah Abdolla	Up to 4 Year Service	20/03/2016	☐
3	Cheryl Brandw	Up to 4 Year Service	25/03/2016	☐
4	Chris McCoy	Up to 4 Year Service	25/03/2016	☐
5	David McGoval	Up to 4 Year Service	25/03/2016	☐
6	Elizabeth Dawl	Up to 4 Year Service	25/03/2016	☐
9	Gilian Hamilto	Up to 4 Year Service	25/03/2016	☐
10	James Beagin	Up to 4 Year Service	25/03/2016	☐
11	Jess Brewer	Up to 4 Year Service	25/03/2016	☐
12	Joanne Myers	Up to 4 Year Service	25/03/2016	☐
14	Melanie Jan Jo	Up to 4 Year Service	25/03/2016	☐
16	Margaret Aspe	Up to 4 Year Service	25/03/2016	☐
17	Peter Markow	Up to 4 Year Service	25/03/2016	☐
18	Raymond Butc	Up to 4 Year Service	25/03/2016	☒
19	Sara Kennedy	Up to 4 Year Service	25/03/2016	☐

A section of the Volunteers table with sensitive information blurred

	ID	FullName	Address	Phone Number	Mobile Number	Email Address	Date Started	Active	Up to 4 Years	5 - 9 Years	10 - 19 Years	20 - 24 Years	Manager	Occasional
	2	Chris McCoy					01/05/2013	☒	☒	☐	☐	☐	☒	☐
	3	Steph Stather					01/01/2012	☒	☐	☒	☐	☐	☒	☐
	4	Jess Brewer					01/08/2014	☒	☒	☐	☐	☐	☐	☐
	5	Donna Bailey					01/09/2012	☒	☐	☐	☐	☐	☐	☐
	6	Gillian Hamilton					01/06/2012	☒	☐	☐	☐	☐	☐	☐
	8	Shirley Park					01/08/2013	☒	☒	☐	☐	☐	☐	☐
	11	Afrah Abdolla					01/08/2014	☒	☒	☐	☐	☐	☐	☐
	13	Cheryl Brandwood					01/07/2012	☒	☒	☐	☐	☐	☐	☐
	14	Christopher Demullet					01/10/2009	☒	☐	☒	☐	☐	☐	☐
	15	David McGowan					01/05/2013	☒	☒	☐	☐	☐	☐	☐
	16	Elizabeth Dawn Sydall					01/03/2014	☒	☒	☐	☐	☐	☐	☐
	18	Georgia Bradley					01/04/2014	☒	☐	☐	☐	☐	☐	☐
	19	Isabel Mary Jackson					01/06/2011	☒	☐	☒	☐	☐	☐	☐
	20	James Reagan					01/12/2014	☒	☒	☐	☐	☐	☐	☐
	21	Janet Marjorie Lythgoe					01/09/2015	☒	☐	☐	☐	☐	☐	☐
	22	Joan Blunsdon					01/11/2011	☒	☒	☐	☐	☐	☐	☐
	23	Joanne Myers					01/06/2012	☒	☒	☐	☐	☐	☐	☐
	24	John Milroy Sloan					01/03/1997	☒	☐	☐	☒	☒	☒	☐

Volunteer training table

[illegible]

Paper register

⑦

DATE	NAME	TIME IN	TIME OUT
12/4/18	Julie		
	Steph		
	Jackie		
	Chas		
	Joss		
	Dave		
	Jayne		
	Jim Barlow	388021	07954783401
	Sandra		
	Edith		
	Muriel		
	Lynne		
	John		
	Jim		
	Brian		
	Bill		
19/4/18	Steph		
	Joss		
	Jackie		
	Julie		
	Jim Barlow ✓		
	David		
	Edith		
	Sandra		
	Neil		
	Brian		122.05
	Jim		
	Dorothy		
	Bill		

Computer club questions

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1 is Strongly Disagree 3 is Neutral 5 is Strongly Agree

*Required

1. It would be useful to be able to create forms and myself by selecting questions from a list *

Mark only one oval.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☒	Strongly Agree

2. There is a wide number and range of forms and questions available for me to practice *

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Strongly Disagree	☐	☐	☐	☒	☐	Strongly Agree

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6. The current system is confusing to use *

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Strongly Disagree	☐	☐	☒	☐	☐	Strongly Agree

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Computer club questions

The system refers to the current method you use to fill in forms. This is both learnmyway.com and the word documents.

1 is Strongly Disagree 3 is Neutral 5 is Strongly Agree

*Required

1. It would be useful to be able to create forms and myself by selecting questions from a list *

Mark only one oval.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☒	☐	☐	Strongly Agree

2. There is a wide number and range of forms and questions available for me to practice *

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	1	2	3	4	5	
Strongly Disagree	☐	☐	☒	☐	☐	Strongly Agree

3. I can easily search for and find a form to practice in the system *

Mark only one oval.

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Strongly Disagree	☐	☐	☐	☒	☐	Strongly Agree

4. I can easily see where I've made an error filling in a form because the system tells me and highlights the error *

Mark only one oval.

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Strongly Disagree	☐	☒	☐	☐	☐	Strongly Agree

5. It would be useful to see what questions a commonly struggle on and which ones take me the longest to complete *

Mark only one oval.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☒	☐	Strongly Agree

6. The current system is confusing to use *

Mark only one oval.

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Strongly Disagree	☐	☐	☐	☒	☐	Strongly Agree

7. A help menu which explains how to use each function of the system would be useful *

Mark only one oval.

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Strongly Disagree	☐	☒	☐	☒	☐	Strongly Agree

8. The forms I've practiced are realistic and are similar to the forms that I've encountered in the real world. (Only answer if you've completed a real world form)

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